

AWS Project Guide: Flask Login + Rectangle Calculator

Project Overview

- This workspace contains two Flask applications: a login service and a rectangle calculator service.
- The login service is under the logic/ folder and runs on port 5000.
- The calculator service is under rectangle_app/ and runs on port 5001.
- The login app redirects to the calculator app after a successful sign-in.

Prerequisites

- Python 3 installed locally.
- pip available for package installation.
- Git configured and connected to the remote repository.
- AWS EC2 instance with SSH access and security groups allowing TCP ports 22, 5000, and 5001.

Local Development: Run the apps

- Open a terminal in the repository root: `c:\aws-project`.
- Run the login service: `cd logic && python app.py`
- Open a separate terminal and run the calculator service: `cd rectangle_app && python app.py`
- Visit the login app at `http://localhost:5000/login`.
- Use username admin and password admin to sign in.
- After login, the app redirects to `http://localhost:5001/home`.

Git Workflow

- Stage changes: `git add .`
- Commit with a message: `git commit -m "Your message"`
- Push to origin main: `git push origin main`
- Make sure the repo URL is correct if deploying from a different machine.

AWS Deployment Steps

- SSH into the EC2 instance using your private key: `ssh -i path/to/key.pem ec2-user@ec2-ip-address`.
- Clone or pull the repo into the remote path, for example `~/cloudrun`.
- Install Python and dependencies on EC2 if needed.
- Start the login app on port 5000 with `python3 logic/app.py`.
- Start the rectangle app on port 5001 with `python3 rectangle_app/app.py`.
- Use `nohup` or `tmux/screen` to keep processes running after logout.
- Confirm the instance security group allows inbound TCP on 5000 and 5001.

AWS Troubleshooting

- If port 5001 is already in use, identify the process: `sudo ss -tulpn | grep :5001`.
- Stop an existing process on the port: `sudo kill` or `sudo pkill -f "python3 rectangle_app/app.py"`.
- Check logs in the remote folder, e.g. `logic.log` and `rectangle_app.log`.
- Confirm the app can be reached from the browser using the EC2 public IP and port.

Notes

- The login app in logic/ now redirects to the calculator app on port 5001 after successful login.
- The calculator app /logout route redirects back to the login app on port 5000.
- UI changes are handled in templates under logic/templates and rectangle_app/templates.